MAT 201 - MATHEMATICAL METHODS I

LECTURE 6: REAL-VALUED FUNCTIONS OF TWO OR THREE VARIABLES

LECTURE C24-1-6 :: MON., JAN. 20, 2025 8:00 AM (WAT) - FRI., JAN. 24, 2025 6:00 PM (WAT)

Prepared by Dr. Ashiribo Senapon WUSU

Email: ashiribo.wusu@lasu.edu.ng | Phone: 08026479471

Mathematics, Lagos State University.

0:01

FUNCTIONS OF TWO VARIABLES

The definition of a function of two variables is very similar to the definition for a function of one variable. The main difference is that, instead of mapping values of one variable to values of another variable, we map ordered pairs of variables to another variable.

0:01

Definition: function of two variables: A function of two variables $z = f(x, y)$ maps each ordered pair (x, y) in a subset D of the real plane $\mathbb{R}^2$ to a unique real number z. The set D is called the domain of the function. The range of f is the set of all real numbers z that has at least one ordered pair $(x, y) \in D$ such that $f(x, y) = z$.

0:01

DOMAINS AND RANGES FOR FUNCTIONS OF TWO VARIABLES

$f(x, y) = 3x + 5y + 2$
This is an example of a linear function in two variables. There are no values or combinations of x and y that cause $f(x, y)$ to be undefined, so the domain of f is $\mathbb{R}^2$. To determine the range, first pick a value for z. We need to find a solution to the equation $f(x, y) = z$, or $3x - 5y + 2 = z$. One such solution can be obtained by first setting $y = 0$, which yields the equation $3x + 2 = z$. The solution to this equation is $x = (z - 2)/3$, which gives the ordered pair $(\frac{z-2}{3}, 0)$ as a solution to the equation $f(x, y) = z$ for any value of z. Therefore, the range of the function is all real numbers, or $\mathbb{R}$.

0:01

DOMAINS AND RANGES FOR FUNCTIONS OF TWO VARIABLES

For the function $g(x, y) = \sqrt{9 - x^2 - y^2}$ to have a real value, the quantity under the square root must be nonnegative:
$9 - x^2 - y^2 \geq 0.$
This inequality can be written in the form
$x^2 + y^2 \leq 9$
Therefore, the domain of $g(x, y)$ is
$\left\{(x, y) \in \mathbb{R}^2 | x^2 + y^2 \leq 9\right\}$

0:01

FUNCTIONS OF TWO VARIABLES

Definition: Vertical Traces: Consider a function $z = f(x, y)$ with domain $D \subset \mathbb{R}^2$. A vertical trace of the function can be either the set of points that solves the equation $f(a, y) = z$ for a given constant $x = a$ or $f(x, b) = z$ for a given constant $y = b$.

0:01

GRAPHING FUNCTIONS OF TWO VARIABLES

Suppose we wish to graph the function $z = f(x, y)$. This function has two independent variables (x and y) and one dependent variable (z). When graphing a function $y = f(x)$ of one variable, we use the Cartesian plane. We are able to graph any ordered pair (x,y) in the plane, and every point in the plane has an ordered pair (x, y) associated with it. With a function of two variables , each ordered pair (x, y) in the domain of the function is mapped to a real number z. Therefore, the graph of the function f consists of ordered triples (x, y, z). The graph of a function $z = f(x, y)$ of two variables is called a surface.

0:01

FUNCTIONS OF MORE THAN TWO VARIABLES

It is useful to take a brief look at functions of more than two variables.

- $\underbrace{f(x,y,z) = x^2 - 2xy + y^2 + 3yz - z^2 + 4x - 2y + 3x - 6}_{\text{a polynomial in three variables}}$
- $g(x,y,t) = (x^2 - 4xy + y^2)\sin t - (3x + 5y)\cos t$

In the first function, (x, y, z) represents a point in space, and the function f maps each point in space to a fourth quantity, such as temperature or wind speed. In the second function, (x, y) can represent a point in the plane, and t can represent time. The function might map a point in the plane to a third quantity (for example, pressure) at a given time t.

0:01

FUNCTIONS OF THREE VARIABLES

Definition: Level surface: Given a function $f(x, y, z)$ and a number c in the range of f, a level surface of a function of three variables is defined to be the set of points satisfying the equation $f(x, y, z) = c$.

0:01

ASSIGNMENT

- Find the equation of the level surface of the function $g(x, y, z) = x^2 + y^2 + z^2 - 2x + 4y - 6z$ corresponding to $c = 2$, and describe the surface, if possible.
- Find vertical traces for the function $f(x, y) = \sin x \cos y$ corresponding to $x = -\pi/4, 0$, and $\pi/4$, and $y = -\pi/4, 0$, and $\pi/4$

LASU-VLAP

You have to complete this lecture within Mon., Jan. 20, 2025 8:00 AM (WAT) - Fri., Jan. 24, 2025 6:00 PM (WAT) in order to mark attendance.

OK